Laplacians

https://asanchowk.github.io/

1 The Laplacian of a function

In this section we describe the interpretation of the Laplacian of a function as the divergence of the gradient of the function.

We start by explicitly defining it as such, and proceeding to parse it after:

Definition 1.1. (Laplacian of f) Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice-differentiable function. Then the Laplacian of f, denoted by $\Delta f : \mathbb{R}^n \to \mathbb{R}$ is the divergence of the gradient of f, denoted by

$$\Delta f = \nabla \cdot \nabla f$$

The fact that f needs to be a twice-differentiable function is more explicit with the following form of the Laplacian of f:

$$\Delta f = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2} f \tag{1}$$

Note that the above definition is for the Laplacian of f, which is different from the Laplace operator itself:

$$\Delta : C^k(\mathbb{R}^n) \to C^{k-2}(\mathbb{R}^n)$$

Parsing Definition (1.1) requires that we first understand the concepts of gradient and divergence first.

Definition 1.2. (Gradient of f) Let $U \in \mathbb{R}^n$ be an open set and $f : U \to \mathbb{R}$ be a C^1 function. Then the gradient of f, denoted as $\nabla f : \mathbb{R}^n \to \vec{\mathbb{R}}^n$ is the function

$$\text{grad } f = \nabla f = \begin{bmatrix} D_1 f \\ D_2 f \\ \dots \\ D_n f \end{bmatrix}$$

In the above definition, $D_i f$ is the partial derivative of f in the i'th standard direction.

Importantly, note that ∇f is a *function*, and if it feels easier, you could write it as the function composition $(\nabla \circ f)(\cdot)$ to make it more obvious. ∇f takes as input a point in $\mathbb{R}^n$ and returns a vector in $\mathbb{R}^n$, which I have tried to make explicit in the definition above by the notation $\nabla f : \mathbb{R}^n \to \vec{\mathbb{R}}^n$.

Again, be careful that this is different than the gradient operator, which is the function:

$$\nabla : C^k(\mathbb{R}^n) \to C^{k-1}(\mathbb{R}^n)$$

defined by

$$\nabla = \begin{bmatrix} D_1 \\ D_2 \\ \dots \\ D_n \end{bmatrix}$$

There are two points I wish to stress here.

Firstly, this makes the gradient of a function *a vector field.* [1]

Secondly, note that this is different from the gradient of f at a point x, which is a *vector*: [2]

Definition 1.5. (Gradient of f at a point) Let $U \in \mathbb{R}^n$ be an open set and $f : U \to \mathbb{R}$ be a C^1 function. Then the gradient of f at the point $x \in U$ is the vector

$$\nabla f(x) = \begin{bmatrix} D_1 f(x) \\ D_2 f(x) \\ \dots \\ D_n f(x) \end{bmatrix}$$

The gradient of f at point x is the vector that denotes the direction of steepest increase of f at x. The entries of $\nabla f(x)$ are the the directional derivatives of f at x in each of the standard directions.

Let us proceed to describe the divergence of a vector field:

Definition 1.6. (Divergence) Let $U \in \mathbb{R}^n$ be an open set, $f : U \to \mathbb{R}$ be a C^1 function and F be a vector field on U. Then the divergence of F is

$$\text{div } F = \nabla \cdot F = \begin{bmatrix} D_1 \\ D_2 \\ \dots \\ D_n \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \\ \dots \\ F_n \end{bmatrix} = D_1 F_1 + D_2 F_2 + \dots + D_n F_n$$

div F is a *function* that returns a scalar value. The function takes in a point $x \in \mathbb{R}^n$ and returns the net outward flow at x.

I like the following description of divergence from Wikipedia:

[1] A vector field is a special type of vector function:

Definition 1.3. (Vector function) A vector function $f : \mathbb{R}^n \to \mathbb{R}^m$ is simply a function that takes in a point in $\mathbb{R}^n$ and returns a vector in $\mathbb{R}^m$.

Contrast that to a scalar function:

Definition 1.4. (Scalar function) A scalar function $f : \mathbb{R}^n \to \mathbb{R}$ is a function that takes in a point in $\mathbb{R}^n$ and returns a real-valued scalar.

A vector field is simply a special type of vector function, where the range is a subset of the domain. This means that the dimension of the domain and range of a vector field is the same. Vector fields are commonly visualized as a collection of arrows, as is done in its Wikipedia page. However, I have found that this might encourage the confusion that vector fields are sets of vectors, instead of a function over a vector space that returns a vector for each point.

Consider a pot full of water that is being heated from underneath, and the heat is making the water move around. The velocity of the water is a vector field, because each point in the pot has a velocity vector associated with it. The temperature of the water is a scalar function, with each point in the pot having a real-valued number denoting the temperature at that point.

[2] This was particularly confusing to me at first, because I had always thought of the gradient of f as a vector, and did not realize that what I was thinking of was actually the definition of the gradient of f at a point. I think this confusion can be alleviated by explicitly mentioning "the gradient of f" vs "the gradient of f at x".

In physical terms, the divergence of a vector field is the extent to which the vector field flux behaves like a source or a sink at a given point. It is a local measure of its "outgoingness" – the extent to which there are more of the field vectors exiting from an infinitesimal region of space than entering it. A point at which the flux is outgoing has positive divergence, and is often called a "source" of the field. A point at which the flux is directed inward has negative divergence, and is often called a "sink" of the field.

Armed with Definition (1.6) and Definition (1.2) we are finally ready to state re-state Definition (1.1):

$$\Delta f = \nabla \cdot \nabla f = \begin{bmatrix} D_1 \\ D_2 \\ \dots \\ D_n \end{bmatrix} \cdot \begin{bmatrix} D_1 f \\ D_2 f \\ \dots \\ D_n f \end{bmatrix} = D_1^2 f + D_2^2 f + \dots D_n^2 f = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2} f$$

Δf is a function that takes as input a point in $\mathbb{R}^n$ and returns a scalar. $\Delta f(x)$ can be interpreted as the net flow out of x.

2 The Graph Laplacian

The last section defined the gradient, divergence and the laplacian in a continuous setting. In this section, we are going to present analogous definitions in the graph setting. Our focus will be more on developing an intuition for the graph Laplacian. Throughout, we will be making references to the definitions of the last section to see how they translate over.

Specifically, we will be working with a directed graph.

2.1 The setup

Consider a two dimensional plane with the standard basis in $\mathbb{R}^2$ with a function f over the plane. Next, discretize the continuous $\mathbb{R}^2$ plane into a graph in $\mathbb{Z}^2$ plane, with the discrete points as nodes and we put an edge if the points are next to each other. This leaves us with a tidy square grid mesh in $\mathbb{Z}^2$, where each node has the same degree (4). General graphs are messier than this grid, so imagine that instead of the nodes being regularly spaced apart the spacing is more erratic, and specifically the degree of each graph may be different. The following picture illustrates this discretization:

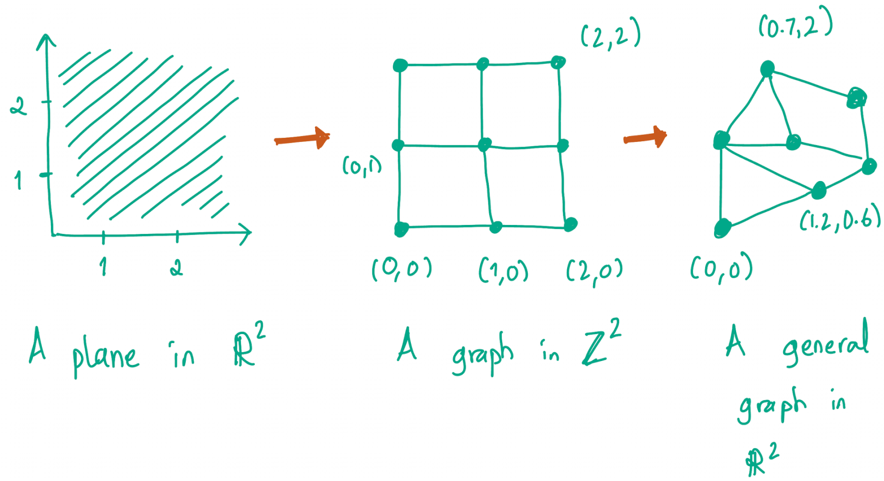


Figure 1: Viewing a general graph as a discretization of a plane. The intermediate step of turning it into a grid graph in $\mathbb{Z}^2$ is not actually necessary.

We can now see that a function f over a plane in $\mathbb{R}^2$ now becomes a function over the nodes of this graph:

$$f : V \to \mathbb{R}$$

2.2 The gradient of a function on a graph

We next establish the notion of the gradient of f in this setting. Recall the definition of the gradient of f at a point (Definition (1.5)), which we re-state below:

(Gradient of f at a point) Let $U \in \mathbb{R}^n$ be an open set and $f : U \to \mathbb{R}$ be a C^1 function. Then the gradient of f at the point $x \in U$ is the vector

$$\nabla f(x) = \begin{bmatrix} D_1 f(x) \\ D_2 f(x) \\ \dots \\ D_n f(x) \end{bmatrix}$$

The entries of $\nabla f(x)$ are the the directional derivatives of f at x in each of the standard directions.

As we saw earlier, the general graph structure sort of destroys the idea of standard directions. However, we can still retain the idea of directional derivatives. For any node of a directed graph, the directed edges associated with the node can be thought of as the directional derivatives in the direction of its neighbors.

Recall the definition of the directional derivative:

Definition 2.1. (Directional derivative) Let $U \subset \mathbb{R}^n$ be an open set, and let $f : U \to \mathbb{R}^n$ be a function. The directional derivative of f at $a \in U$ in the direction of vector $\vec{\mathbf{v}}$ is

$$D_{\vec{\mathbf{v}}} f(a) = \lim_{h \to 0} \frac{f(a + h\vec{\mathbf{v}}) - f(a)}{h}$$

The directional derivative can be thought of as the partial derivative, only in the direction of $\vec{\mathbf{v}}$ instead of a standard direction.

In the graph setting, the equivalent definition is

Definition 2.2. (Directional difference over nodes in a Graph) Let $G = (V, E)$ be a directed graph, and let $f : V \to \mathbb{R}^n$ be a function over the nodes. Let $a, b \in V$ such that there is a directed edge e from a to b, i.e $e = a \to b$.
The directional derivative of f at a in the direction of edge e is

$$D_e f(a) = f(b) - f(a)$$

We are now ready to state our definition for the gradient in a graph setting:

Definition 2.3. (Gradient of f at a node) Let $G = (V, E)$ be a directed graph and $f : V \to \mathbb{R}$ be a function. Let $A \in V$ and let A have k neighbors i.e $\deg(A) = k$ and let $e_1, \dots, e_k$ be the edges connecting A to these neighbors. Then the gradient of f at A is the vector

$$\nabla f(A) = \begin{bmatrix} D_{e_1} f(A) \\ D_{e_2} f(A) \\ \dots \\ D_{e_k} f(A) \end{bmatrix}$$

This is the graph-theoretic analogy to Definition (1.5). [3]

[3] An interesting thing to note about this definition is that the length of the gradient vector differs from node to node depending on the node's degree. This is different than when we were in the continuous case of Definition (1.5). This is because we have lost the orientation of the "standard" directions that every point in $\mathbb{R}^n$ has. A point in $\mathbb{R}^n$ has n (linearly independent) directions it can make a movement towards, so it has n directional derivatives and therefore its gradient is a vector of length n. Similarly, a node A of a graph, which is the graph-theoretic analogy for a point, can move in $\deg(A)$ directions, so it has $\deg(A)$ directional differences and therefore its gradient vector is of length $\deg(A)$.

Consider the directed edge $e : A \to B$. Since the edge is directed,

$$D_e f(A) = D_e f(B) = f(B) - f(A)$$

Because of this it follows that the directional difference is more a property of the edge e than of the nodes it connects, i.e

$$D_e f(A) = D_e f(B) = D_e f$$

This allows us to build the notion of gradient of f on a graph, which is the graph analogue of Definition (1.2):

Definition 2.4. (Gradient of f on a Graph) Let $G = (V, E)$ be a directed graph and $f : V \to \mathbb{R}$ be a function. Let $|E| = n$. Then the gradient of f on G, denoted as $\nabla f : V \to \mathbb{R}$ is the function

$$\text{grad } f = \nabla f = \begin{bmatrix} D_{e_1} f \\ D_{e_2} f \\ \dots \\ D_{e_n} f \end{bmatrix}$$

2.3 The divergence of the gradient of a function on a graph

Recall the definition of the Laplacian of the graph in Definition (1.1), which states:

$$\Delta f = \nabla \cdot \nabla f$$

While this is the common notation, for our purposes of making clear what is happening here we will name the two ∇s specifically, and rename the above expression to be

$$\Delta f = \nabla_2 \cdot \nabla_1 f$$

We will use this notation to interpret these operators both in the continuous setting and the graph setting:

The Continuous Setting	**The Graph Setting**
$f : \mathbb{R}^n \to \mathbb{R}$ is a function over the points in $\mathbb{R}^n$.	$f : V \to \mathbb{R}$ is a function over the nodes.
$\nabla_1 f$ is a function that takes in a point in $\mathbb{R}^n$ and returns a vector of directional derivatives (i.e the gradient) at that point.	$\nabla_1 f$ is a function that takes in a node and returns a vector of directional differences (which are associated with edges) at that point.
∇_2 takes in that vector of directional derivatives and returns a scalar value associated with the original point.	∇_2 takes in that vector of directional differences, and returns a scalar value associated with the original node.

There are two important things to note here:

First: for the continuous setting, both ∇_1 and ∇_2 are operators that take in a function that takes an input in $\mathbb{R}^n$. ∇_1 takes as input a function that takes in points in $\mathbb{R}^n$, and ∇_2 takes as input the gradient of a function, which is a vector in $\mathbb{R}^n$. Contrast this to the graph setting, where ∇_1 takes in a function over the nodes, and ∇_2 takes in a function over the edges. This is why in this setting I think it is helpful to differentiate between the ∇_1 and ∇_2. The graph setting is, in this sense, more general than the continuous setting.

Second: For the graph setting, the fact that ∇_2 and ∇_1 are operating on functions over edges and nodes respectively has an immediate consequence - that the idea of "directional difference" is different for them.

The notion of the directional difference over nodes is captured by our Definition (2.2) already. We need an additional definition of difference over the edges, which we will do now:

Definition 2.5. (Directional difference over edges in a Graph) Let $G = (V, E)$ be a directed graph, and let $g : E \to \mathbb{R}$ be a function over the edges. For any $v \in V$, let $\mathcal{O}$ be the set of edge incident on v such that the edges in $\mathcal{O}$ are directed outwards from v. Similarly, let $\mathcal{I}$ be the set of edge incident on v such that the edges in $\mathcal{I}$ are directed inwards from v. Then, the directional derivative of g at v is

$$Dg(v) = \sum_{e \in \mathcal{O}} g(e) - \sum_{e \in \mathcal{I}} g(e)$$

In other words, it is the net outwards value of g at the node v. Equivalently, this is the *divergence* of g at the node.

This is this directional difference operator used by ∇_2 in the graph setting.

Note that we had to deal with defining two notions of difference (Definition (2.2) and Definition (2.5)) for the graph setting, but the continuous setting made do with the same notion - you could simply take the partial derivatives both times. Again, this is because the graph setting is, atleast in this sense, more general than the continuous setting.

2.4 An example to see that $L = D - A$

Across computer science, the common way that the Laplacian matrix is represented is

$$L = D - A$$

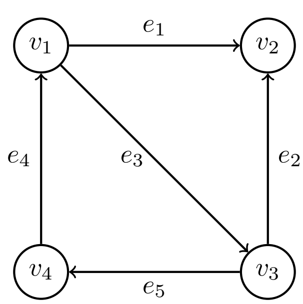
where D and A are the degree and adjacency matrices respectively. We will work through an example to see how this formula comes to be.

More specifically, we will actually work an example to show

$$L = A - D$$

instead, and then talk about why the difference in sign is a matter of convention.

This is the directed graph we will use as our working example:



Following Definition (2.4), the gradient of f on this graph is:

$$\nabla f = \begin{bmatrix} D_{e_1} f \\ D_{e_2} f \\ D_{e_3} f \\ D_{e_4} f \\ D_{e_5} f \end{bmatrix} = \begin{bmatrix} f(v_2) - f(v_1) \\ f(v_2) - f(v_3) \\ f(v_3) - f(v_1) \\ f(v_1) - f(v_4) \\ f(v_4) - f(v_3) \end{bmatrix}$$

We start by asking what matrix J we can left-multiply f with to get ∇f:

$$\underbrace{\begin{bmatrix} ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}}_{J} \begin{bmatrix} f(v_1) \\ f(v_2) \\ f(v_3) \\ f(v_4) \end{bmatrix} = \underbrace{\begin{bmatrix} f(v_2) - f(v_1) \\ f(v_2) - f(v_3) \\ f(v_3) - f(v_1) \\ f(v_1) - f(v_4) \\ f(v_4) - f(v_3) \end{bmatrix}}_{\nabla f}$$

(We know that $J \in \mathbb{R}^{5\times 4}$ to get the dimensions correct for this matrix multiplication.)

If you try to work this out, hopefully it is clear that the matrix J that performs this mapping is:

$$\underbrace{\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}}_{J} \begin{bmatrix} f(v_1) \\ f(v_2) \\ f(v_3) \\ f(v_4) \end{bmatrix} = \underbrace{\begin{bmatrix} f(v_2) - f(v_1) \\ f(v_2) - f(v_3) \\ f(v_3) - f(v_1) \\ f(v_1) - f(v_4) \\ f(v_4) - f(v_3) \end{bmatrix}}_{\nabla f}$$

We now proceed to compute the divergence of ∇f. Recall that we described divergence as the local measure of the "outgoingness" of a point/node. Using Definition (2.5), the divergence vector is

$$\begin{bmatrix} D_{e_1} f + D_{e_3} f - D_{e_4} f \\ -D_{e_1} f - D_{e_2} f \\ D_{e_2} f + D_{e_5} f - D_{e_3} f \\ D_{e_4} f - D_{e_5} f \end{bmatrix}$$

Again we ask, what matrix K applied to ∇f gives us the above vector?

$$\underbrace{\begin{bmatrix} ? & ? & ? & ? & ? \\ ? & ? & ? & ? & ? \\ ? & ? & ? & ? & ? \\ ? & ? & ? & ? & ? \end{bmatrix}}_{K} \underbrace{\begin{bmatrix} D_{e_1} f \\ D_{e_2} f \\ D_{e_3} f \\ D_{e_4} f \\ D_{e_5} f \end{bmatrix}}_{\nabla f} = \underbrace{\begin{bmatrix} D_{e_1} f + D_{e_3} f - D_{e_4} f \\ -D_{e_1} f - D_{e_2} f \\ D_{e_2} f + D_{e_5} f - D_{e_3} f \\ D_{e_4} f - D_{e_5} f \end{bmatrix}}_{\Delta f}$$

Again, we know the dimensions of K because of the dimensions of ∇f and Δf. We see that the following is the result:

$$\underbrace{\begin{bmatrix} 1 & 0 & 1 & -1 & 0 \\ -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}}_{K} \underbrace{\begin{bmatrix} D_{e_1} f \\ D_{e_2} f \\ D_{e_3} f \\ D_{e_4} f \\ D_{e_5} f \end{bmatrix}}_{\nabla f} = \underbrace{\begin{bmatrix} D_{e_1} f + D_{e_3} f - D_{e_4} f \\ -D_{e_1} f - D_{e_2} f \\ D_{e_2} f + D_{e_5} f - D_{e_3} f \\ D_{e_4} f - D_{e_5} f \end{bmatrix}}_{\Delta f}$$

Finally, we put it all together to determine that the Laplacian of f is the transformation of f under the operation $K \circ J$:

$$\underbrace{\begin{bmatrix} 1 & 0 & 1 & -1 & 0 \\ -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}}_{K} \underbrace{\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}}_{J} \begin{bmatrix} f(v_1) \\ f(v_2) \\ f(v_3) \\ f(v_4) \end{bmatrix} = \underbrace{\begin{bmatrix} -3 & 1 & 1 & 1 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & 1 \\ 1 & 0 & 1 & -2 \end{bmatrix}}_{\Delta} \begin{bmatrix} f(v_1) \\ f(v_2) \\ f(v_3) \\ f(v_4) \end{bmatrix}$$

Notice that in the above expression, $\Delta = L = A - D$:

$$\underbrace{\begin{bmatrix} -3 & 1 & 1 & 1 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & 1 \\ 1 & 0 & 1 & -2 \end{bmatrix}}_{\Delta} = \underbrace{\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}}_{A} - \underbrace{\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}}_{D}$$

Remark 2.1. Δ is the Laplacian operator, not the Laplacian of f.

The reason that we have $L = A - D$ and not $L = D - A$ is because in Definition (2.2) we arbitrarily defined the directional difference of a directed edge $e : a \to b$ to be

$$D_e f(a) = f(b) - f(a)$$

If we had chosen to instead define it as

$$D_e f(a) = f(a) - f(b)$$

then we would have obtained $L = D - A$.

This arbitrariness is the reason we see Laplacians defined with either the positive and negative signs all over the place.

2.5 Some miscellaneous remarks

1. One might find the following very bizarre at first (like I did):

   One representation of the Laplacian is

   $$\Delta f = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2} f$$

   which is a sum of the second derivatives. Another representation is something like Definition (2.5) which is essentially a sum of first derivatives. How are these two representations saying the same thing?

   One (direct) way is to see, through our discussion of the divergence in the graph setting, that the summing of the first derivatives is a form of differencing.

   If you wish to go deeper with the math, Justin Solomon recommends (in his Laplacians lecture in his Shape Analysis 2023 lecture series on Youtube) to google "strong from and weak form differential equations".

2. ∇_1 and ∇_2 are adjoints to each other.

3. It is remarkable that $KJ = M^T M$, where M is the incidence matrix. I would like to explore the fact that this is a symmetric matrix, and has connections to the least squares problem.